

# A Single-Chain Implicit DNA Model Predicts H3 Histone Tail Phase-Separation Propensity from Charged-Residue Positioning

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## Abstract

The patterns of charged residues and the net charge of the histone H3 tail affect nucleosome condensate formation and dynamics, as shown by explicit DNA models. However, such models are very computationally expensive and slow, despite their accuracy. Therefore, the question here is whether a much simpler and cheaper model can still accurately predict the relative phase-separation behavior of different H3 sequences. In this paper, a coarse-grained Monte Carlo model of the H3 tail is developed which uses Ashbaugh-Hatch interactions with CALVADOS 2 parameters, Debye-Hückel electrostatics, and an implicit nucleosome model represented by a charged sphere. The model tested nine H3 variants with known saturation concentrations ( $c_{\text{sat}}$ ). The simulations showed that the whole-chain radius of gyration ( $R_g$ ) didn't have a statistically significant correlation with  $c_{\text{sat}}$ . On the contrary, the mean distance of the charged residues from the nucleosome surface showed a strong positive correlation with the  $c_{\text{sat}}$  values across various DNA attraction strengths, with Spearman  $\rho = 0.75 - 0.87$  and  $p < 0.05$ . These results reflect that a simpler single-chain, implicit DNA model can capture the relative phase-separation behavior of different H3 sequences. The model works best when the interaction of the charged residue with DNA is tracked rather than the whole chain.

## Introduction

Nucleosomes are the primary building blocks of chromatin, consisting of a nucleosome core, linker DNA, and linker histone. The nucleosome core is conserved throughout eukaryotic cells and contains eight histones, two of each H2A, H2B, H3, and H4 histone proteins [1]. H3 and H4 form a tetramer together, and H2A and H2B histones form two dimers that are attached to the tetramer [2]. The 146 bp DNA molecule wraps around the histone octamer 1.65 times in the left-hand direction. The dyad (the central point) of the nucleosome divides the DNA into 73 and 72 bp sections. This dyad is marked as the superhelix location (SHL) 0, where the bases upstream are assigned a positive sign, and the downstream bases are assigned a negative sign [1]. Histones have intrinsically disordered regions (IDRs) in N-terminal tails [2] and are highly positively charged due to the excess of Arg and Lys residues [3]. Due to this

nature, the histones are involved in chromatin packaging, accessibility, and nucleosome stability. Therefore, removing or mutating histone tails causes instability and, in H3, increases nucleosome movement and interaction with DNA [2]. According to Struhl (2024), not all IDRs are flexible and disordered regions; some can bind to other proteins or DNA molecules, act as linkers, or form weak bonds to create condensates. Some IDRs would lose their flexible structure and become more rigid when involved in interactions. Other IDRs have rigid structures, or their structures don't reveal their biological functions because whether, when, or how they are rigid or flexible depends on their interactions [4]. Most of the IDPs are polyampholytes, which contain both positively and negatively charged amino acids. The conformation of IDPs depends on the fraction of charged residues (FCR), net charge per residue (NCPR), as well as the position and order of them. The order in which charged residues in IDRs are arranged is expressed by parameter  $\kappa$ , where a low  $\kappa$  value indicates mixed negative and positive residues, and a high  $\kappa$  value indicates that the same charges form clusters. The strong polyampholyte IDRs have lower  $\kappa$  values yet more charged residues overall ( $\text{FCR} > 0.3$ ), making them more sensitive to FCR and  $\kappa$ , forming random coils. These patterns were identified using the ABSINTH force field, and implicit solvents were used for simulations [5].

The chromatin can undergo phase separation at the nucleosome core particle (NCP) level, partially driven by electrostatic interaction between positively charged histone tails and negatively charged DNA molecules. Histone H3 and H4 tails are especially important for such interactions since removing them or neutralizing their charges through post-translational modifications (PTMs) would make the DNA-histone interactions weaker and reduce phase separation. Hammonds et al. (2026) systematically modified the residues in the histone H3 tail to understand how the position, number, and the residue itself can alter nucleosome phase separation. They also modified Arg and Lys to Gln in the terminal and central regions to neutralize their opposite charges, converted Arg residues to Lys, and Lys to Arg [3].

The multi-chain, explicit simulation of DNA is accurate but more computationally expensive. On the other hand, even though the single-chain implicit DNA simulation is significantly faster, it provides a much simpler representation condensate conditions and intermolecular interactions. Therefore, the question is whether this implicit DNA model captures the effects the sequence-dependent mutations have on phase separation enough to rank the H3 tail mutations by their propensity while maintaining its speed. To assess this, we built a coarse-grained Monte Carlo model of an isolated H3 tail at the implicit nucleosome surface using the Ashbaugh-Hatch potential [6] parameterized with CALVADOS 2 [7] and Debye-Hückel electrostatics. This model was tested on nine H3 tail sequence variants obtained from Hammonds et al (2026) [3] and showed that the radius of gyration  $R_g$  of the whole chain and phase separation propensity were not correlated. Nevertheless, the mean distance of the charged residues in the H3 tail from the nucleosome surface was strongly correlated with the phase separation propensity, which remained strong even when the model's parameters were increased by ten times.

## Methods

### System representation

In the coarse-grained model, each H3 tail residue is represented as a single bead, where the neutral and non-ionizable, positively charged, and negatively charged residues are represented by 0, +1, and -1, respectively (His is treated as neutral since it is deprotonated at physiological pH). A CALVADOS 2

hydrophobicity parameter ( $\lambda$ ) and van der Waals radius ( $\alpha$ ) [8] were assigned to each residue and used in an HPS coarse-grained framework [9].

## Bonded interactions

The adjacent beads' interactions with one another are represented by a harmonic spring:

$$V = \frac{1}{2}k(d - d_0)^2$$

where  $k = 507.866 \text{ kT/nm}^2$  and  $d_0 = 0.38 \text{ nm}$  in  $C_\alpha - C_\alpha$  bond.

## Non-bonded interactions

The non-adjacent beads interact by the Ashbaugh-Hatch potential:

$$V_{AH}(r) = V_{LJ}(r) + (1 - \lambda_{ij})\epsilon \text{ for } r \leq 2^{1/6} \cdot \sigma_{ij}$$

$$V_{AH}(r) = \lambda_{ij} \cdot V_{LJ}(r) \text{ for } 2^{1/6} \cdot \sigma_{ij} < r < r_c$$

where  $V_{LJ} = 4\epsilon[(\frac{\sigma_{ij}}{r})^{12} - (\frac{\sigma_{ij}}{r})^6]$ ,  $\lambda_{ij}$  and  $\sigma_{ij}$  are arithmetic means of the pairs'  $\lambda$  and  $\sigma$ ,  $\epsilon = 0.8368 \text{ kJ/mol}$ , and  $r_c = 2.0 \text{ nm}$  at  $298.15 \text{ K}$ . The bond interactions between the adjacent beads are excluded from this formula.

## Electrostatic interactions

The charged beads also interact by Debye-Hückel screened Coulomb potential,

$$V_{elec}(r) = \frac{l_B \cdot q_i \cdot q_j \cdot e^{-\kappa r}}{r},$$

Where Bjerrum length  $l_B = 0.7 \text{ nm}$  and the inverse Debye screening length  $\kappa = 1.2 \text{ nm}^{-1}$  (corresponds to  $\sim 150 \text{ mM}$  monovalent salt).

## Implicit nucleosome core/ DNA surface

The implicit nucleosome core and DNA are represented as a hard wall with a radius of  $R_{core} = 4.25 \text{ nm}$  (which is centered at the origin). The beads at this hard wall ( $< 4.25 \text{ nm}$ ) get an infinite energy penalty. The positively charged beads are further attracted to the nucleosome core, which is a screened-Coulomb interaction with the surface rather than the center of the implicit nucleosome:

$$V_{DNA}(i) = \frac{l_B \cdot q_i \cdot Q_{DNA} \cdot e^{-\kappa d_s}}{d_s},$$

where  $d_s = \max(|r_i| - R_{core}, 0.30 \text{ nm})$  and  $Q_{DNA} = -30$  represent the local DNA/ core surface.

The parameter *dna\_attraction\_scale* is also calibrated as described below.

## Chain initialization

The initial configurations were based on the Markov chain principle, and each bead was positioned at a random orientation with a fixed, specific bond length distance from the previous bead. The bead was resampled if it violated the hard-wall exclusion or was within  $0.85 \times$  the pairwise contact distance  $\frac{a_i + a_j}{2}$  of any of the previous non-bonded beads. This step is important because it otherwise allows for steric overlapping beads and leads to very high and inaccurate energy.

## Monte Carlo sampling

The H3 tail configuration was sampled by single-bead Metropolis Monte Carlo, where in each step, one bead is selected by chance and moved in a random vector with a step size of 0.08 nm. This movement is accepted with a probability of  $\min(1, e^{-\Delta E})$  where the energy  $E$  is expressed in  $k_{BT}$ . The step size was adjusted to give an acceptance rate of 20-50%. Each run consisted of 20000 equilibration steps and 50000 production steps in which the configurations were sampled and recorded every 100 steps. The equilibration was monitored by checking the radius of gyration  $R_g$  at each interval and comparing the halves of the simulation. For each condition, 5 independent replicates with different random seeds were run, and their uncertainty is presented as the standard error across all of the independent replicates.

## Observables

For each configuration, two quantities were calculated: (i) the whole-chain radius of gyration,

$$R_g = \sqrt{\left[\left(\frac{1}{N}\right) \sum (r_i - r_{cm})^2\right]},$$

which describes the overall size of the histone H3 chain, and (ii) the mean distance of only the positively charged beads from the nucleosome core surface (or so referred to as “charge-residue distance”). This factor is important to calculate since most of the amino acids are neutral and averaging over the whole chain can reduce the sequence-dependent signals.

## Validation dataset

The model’s performance was compared to nine variants of the H3(1-44) sequence for which their NCP saturation concentration ( $c_{sat}$ ) was published by Hammonds et al (2026) [3]. Their database consisted of the unmodified sequence, Lys→Gln substitution at terminal (K4, K9) and central (K14, K18, K23, K27) positions, tested individually or combined; Arg→Gln substitutions at terminal (R2, R8) and central (R17, R26) positions, also tested individually or combined; a quadruple substitution of Arg→Lys; and a sextuple substitution of Lys→Arg, along with their saturation concentrations ( $c_{sat}$ ).

## Calibration

The *dna\_attraction\_scale* parameter is used for calibration, for which four values of [0.2, 0.5, 1.0, 2.0] were used, and the Spearman correlation between the simulated charge residue resistance and the  $c_{sat}$

values across all nine runs was calculated. The value with the highest correlation (0.2) was used for the final analysis. Using the unmodified sequence confirmed that this parameter didn't impact the charge-neutralized control simulations, which is expected since its interaction depends on the presence of the charged residues.

## Statistical analysis

Spearman's rank correlation coefficient was used to measure the relationship between the simulation results and the experimental  $c_{\text{sat}}$  values. The statistical significance was defined as  $p < 0.05$ . All analyses were done using the Python NumPy and SciPy packages.

## Results

### Charge-only perturbation of the DNA attraction term does not noticeably change whole-chain condensation.

To distinguish between the effects of implicit DNA attraction and the internal electrostatic forces, the `dna_attraction_scale` parameter was tested under two conditions, while the other interactions and parameters remained constant: (i) calibrated and (ii) set to 0, where it is completely turned off. The parameter was set to 0.2, 0.5, and 1.0; the overall radius of gyration ( $R_g$ ) shows very little difference between active and inactive attraction states (1.31 nm-1.37nm). These variations are similar to the background noise across independent runs (signal-to-noise ratio of 0.31-0.33 across all three scales, based on five independent runs per condition). The Monte Carlo acceptance rates for the control simulations were between 0.28 and 0.31, which falls in the optimal range for phase-space sampling, meaning the absence of structural changes indicated the true physical behavior of the whole-chain  $R_g$ , not algorithm limitations or sampling artifacts.

### The average distance of the charged residues to the core surface is more responsive to DNA attraction than to the whole-chain compaction $R_g$ .

When the distance of the only positively charged residues of the H3 tail to the nucleosome core was measured, there was a clear distinction between when the DNA attraction was on and off across all the scales, where the signal-to-noise ratio was 3.77 - 5.05, which is more than ten times greater than the signal measured from the whole-chain  $R_g$ .

Only 14/44 residues of the histone H3 tail are positively charged. Therefore, the local conformation and  $R_g$  decrease below a detectable threshold across these residues when they are averaged over the whole chain. However, measuring them across the positively charged residues provides a more realistic structural response.

The charged-residue distance correctly ranks the H3 tail mutants based on their calculated  $c_{\text{sat}}$  values.

The charged-residue distance for all nine histone H3 mutants published by Hammonds et al. (2026) [3] was calculated in the validation set using three independent runs per sequence, as described in Table 1. The results showed a strong correlation between the charged-residue distance and the  $c_{\text{sat}}$  values, with Spearman's  $\rho = 0.87$  and  $p = 0.003$  (Fig. 1). The results also showed that the lower the  $c_{\text{sat}}$  values, the shorter the distances of charged residues to the nucleosome core, such as R2/8/17/26K and K4/9/14/18/23/27R mutants. On the other hand, the K4/9/14/18/23/27Q mutant shows higher  $c_{\text{sat}}$  and a longer charged-residue distance. For the nine variants (Fig. 1, Table 1), screening-level simulation settings were used to prevent the model from being computationally expensive. Each sequence was simulated with 8000 equilibration steps, 4000 production steps, and three replicates. The on/off simulation setting was used on more extensive scales with 20000 equilibration steps, 10000 production steps, and five replicates.

The positive correlation between  $c_{\text{sat}}$  values and charged-residue distance is strong across many `dna_attraction_scale` values.

The simulation was run at four different DNA attraction scales: 0.2, 0.5, 1.0, and 2.0, all of which showed a strong positive correlation with  $c_{\text{sat}}$   $\rho = 0.75 - 0.87$  and  $p < 0.05$  (Fig. 2). The `dna_attraction_scale = 0.2` is chosen as the primary scale since it results in the strongest positive correlation and the weakest DNA attraction. Whole-chain  $R_g$  across the sequences showed no significant correlation with  $c_{\text{sat}}$ , where Spearman's  $\rho = -0.19$  and  $p = 0.62$ , which confirms that the correlation explained by the charge-residue distance doesn't apply to whole-chain compaction.

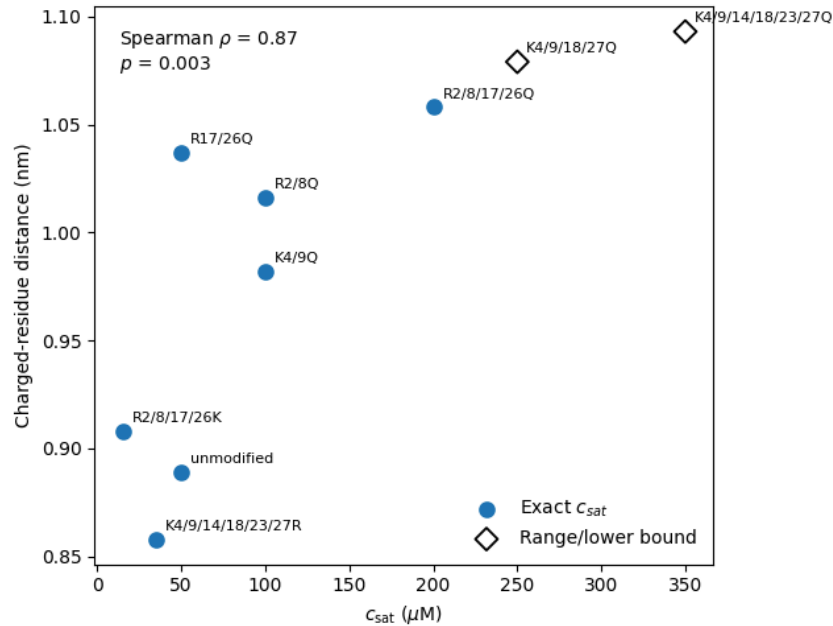
Table 1

*Nine Experimental Variants and Their  $c_{\text{sat}}$  values*

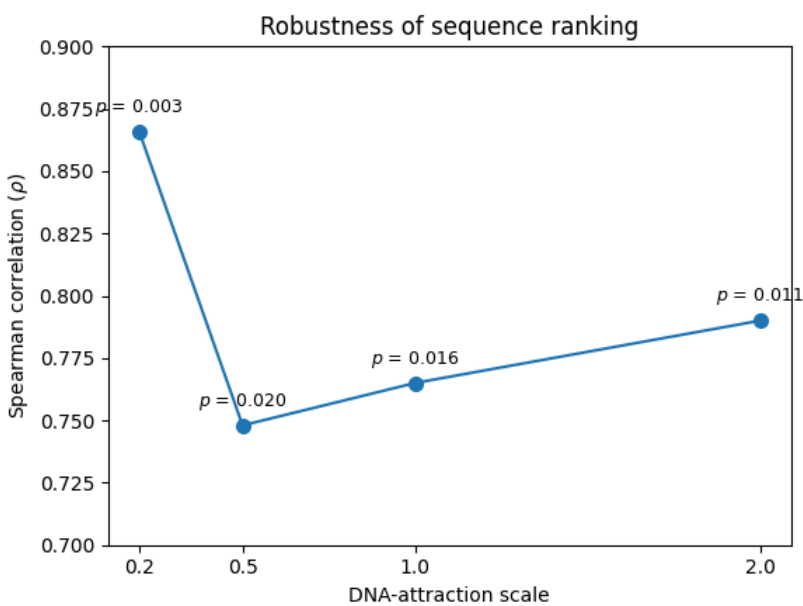
| Mutant            | Description           | $C_{\text{sat}}$ $\mu\text{M}$ | Charged-residue distance (nm) <sup>†</sup> |
|-------------------|-----------------------|--------------------------------|--------------------------------------------|
| unmodified        | WT H3(1-44)           | 50                             | 0.889                                      |
| R2/8Q             | Terminal Arg→Gln (x2) | 100                            | 1.016                                      |
| R17/26Q           | Central Arg→Gln (x2)  | 50                             | 1.037                                      |
| R2/8/17/26Q       | All Arg→Gln (x4)      | 200                            | 1.058                                      |
| K4/9Q             | Terminal Lys→Gln (x2) | 100                            | 0.982                                      |
| K4/9/18/27Q       | Mixed Lys→Gln (x4)    | 200-300 <sup>‡</sup>           | 1.079                                      |
| K4/9/14/18/23/27Q | All Lys→Gln (x6)      | No PS $\leq 300^{\ddagger}$    | 1.093                                      |
| R2/8/17/26K       | All Arg→Lys (x4)      | 15                             | 0.908                                      |

|                   |                  |    |       |
|-------------------|------------------|----|-------|
| K4/9/14/18/23/27R | All Lys→Arg (x6) | 35 | 0.858 |
|-------------------|------------------|----|-------|

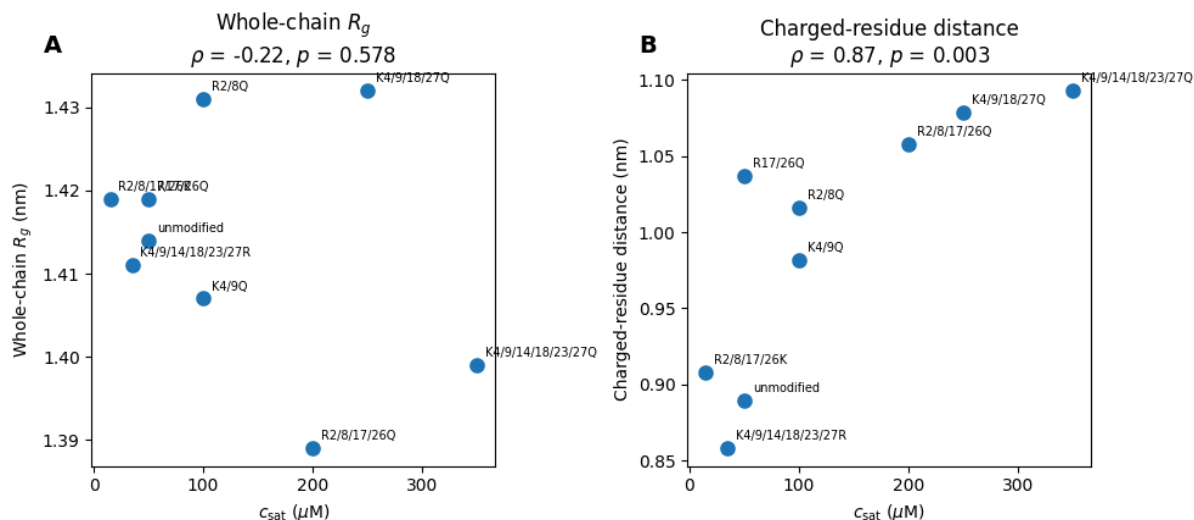
<sup>†</sup> at scale = 0.2, n = 3 replicates. <sup>‡</sup> reported as a range or lower boundary in the source paper, not an exact value, which are marked with diamonds rather than circles in Fig. 1.



**Figure 1.** *Nine H3 tail mutants ranked correctly.* Each point is a mutant. The x-axis shows saturation condensation ( $c_{sat}$ ). The y-axis shows the charged-residue distance in nm. The filled circle nodes are the exact published  $c_{sat}$ , and the diamonds are the approximations. Spearman's  $\rho = 0.87$ ,  $p = 0.003$ .



**Figure 2.** *The correlation is robust across different DNA attraction strengths.* A similar comparison in Figure 1 was extended to four different scales of the DNA attraction strength. All the scales provide a statistically significant strong correlation.



**Figure 3.** *Whether the signal is detectable depends on which observable residue is chosen.* The whole-chain  $R_g$  is not correlated with the variant ranking (A), unlike the charged-residue distance (B).

## Discussion

Only a small fraction of H3 tail residues are affected by DNA attraction since only 14/44 of them are charged, and only 10 of them can be modified to another residue. The whole chain  $R_g$  considers the entire tail. The  $R_g$  is affected by residue interactions, volume exclusion, and chain flexibility. Therefore, the small impacts of the DNA attraction are negligible, and they give a signal-to-noise ratio of  $\sim 0.3$ . The charge-residue distance measures only the residues that are responsive to the DNA attraction, which are more noticeable with a signal-to-noise ratio of 3.8 - 5.1.

Arg and Lys are treated similarly by the model, since they both have a +1 charge and the model cannot differentiate between these residues. However, in the experiments, the Arg residues can stimulate phase separation more strongly than Lys because of cation- $\pi$  interactions and easily losing their hydration shell. These factors are not included in the model. The R2/8/17/26K mutant and the unmodified H3 tail have the same charges because Arg is substituted with Lys at the same residues, even though the  $c_{\text{sat}}$  decreased from 15  $\mu\text{M}$  to 50  $\mu\text{M}$ . However, the charged-residue distance predicted by the model is 0.908 nm in this mutant, which is longer than the one in the unmodified (0.889 nm), which is the opposite of the trends the model suggested. The K4/9/14/18/23/27 R mutant follows the expected pattern since its charge-to-charge distance is shorter (0.858 nm), and its  $c_{\text{sat}}$  is also less (35  $\mu\text{M}$ ). There is an overall strong correlation  $\rho = 0.87$  because most of the differences between the sequences are due to the charge-changing Q substitutions, which are also included by the model. However, since Arg and Lys have the same charges and their substitution doesn't change the overall net charge, the model cannot distinguish them. This is a limitation of the model (that would be amended) because the chemical differences between residues with the same charge are not addressed. To resolve this, Arg and Lys can be given different DNA attraction

strengths, which can be scaled by using their CALVADOS 2  $\lambda$  values with a stronger contribution for Arg to add more residue-specific chemistry without compromising computational cost.

This model doesn't simulate a condensate or explicit DNA; it is faster and simpler. The model was validated against the experimental results and their more detailed simulations. The two models have different approaches and goals; the explicit DNA model can study condensate formation, droplet dynamics, chain interactions, and viscosity, while this implicit model captures the H3 tail dynamics and interactions with an implicit DNA core. This implicit model is, however, much faster.

The model can be used as a pre-screening tool and rank H3 tail variants at low computational cost and high speed. The most promising variant can then be tested with more detailed simulations or experiments. The Spearman correlation showed that the model can rank the sequences but cannot predict the  $c_{\text{sat}}$  values.

## Data and code availability

The code is available at <https://github.com/Aida-Rezaei/H3TailMapper/tree/main>. The force field parameters are from the CALVADOS dataset [7,8]. The validation dataset, sequences, and published  $c_{\text{sat}}$  are obtained from Hammonds et al. (2026) [3].

## References

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